

Ambient Lighting for Improving Passengers' Acceptance of Driverless Buses: Combining the Evaluation in Virtual and Real Environment

Satoru Ogino* Yanbin Wu Toru Kumagai Takahiro Miura Masaki Masuda Koya Takahashi Sota Suzuki Naohisa Hashimoto

National Institute of Advanced Industrial Science and Technology

ABSTRACT

To address traffic issues, the integration of autonomous vehicles, which operate without an onboard driver, is highly anticipated. However, the social acceptance of driverless vehicles is hindered by passengers' apprehension. Various strategies have been explored to alleviate passenger's concern, such as Head-Up Displays (HUDs) that relay information about the vehicle's behavior and status, which generally used the central field of view. In contrast, our research introduces an ambient lighting system that utilizes peripheral vision, liberating passengers from the necessity of focusing their central vision on this information provision and allowing them to seamlessly receive information while engaged in other activities. Our proposed system aims to diminish passengers' anxiety in autonomous buses by delivering crucial information about the vehicle's behavior and status through ambient lighting, which taps into their peripheral vision. This approach enables passengers to stay informed without disrupting their ongoing tasks. To validate the effectiveness of our system, we are conducting experiments in both Virtual Reality (VR) and Real Space, ensuring a comprehensive evaluation. The outcomes of these experiments, combined with relevant research findings, will contribute to the assessment of our proposed ambient lighting system's viability.

Keywords: Autonomous Vehicles, Human Machine Interface, Virtual Reality, In-vehicle Interface.

Index Terms: Autonomous vehicles; Human factors; Human-machine interface; User experience; Virtual reality;

1 INTRODUCTION

In Japan, societal challenges, including a declining birthrate, an aging population, and urban population concentration, raise concerns about escalating issues in road transportation. Amidst this transformative period in the automotive industry, marked by the emergence of CASE (Connected, Autonomous, Shared, Electric) technologies, there is a notable emphasis on the potential of driverless vehicles to address these challenges.

While automated buses are being developed globally, the occurrence of unexpected sudden movements such as accelerations, decelerations, and emergency stops in driverless buses can impose both physical and emotional burdens on passengers. Such uncertainties may impede social acceptance and thus affect the adoption of driverless buses. Our research aims to mitigate this unease by focusing on conveying essential

information about the vehicle's behavior and status to passengers to improve their acceptance.

Traditionally, methods of relaying information through messages and icons require passengers to engage their central vision. By shifting this task to peripheral vision, information can be consistently received, even while passengers are occupied with central vision tasks. Human Machine Interface (HMI) utilizing light color and flashing patterns has proven to be as reliable and effective as voice-based HMI. Therefore, non-verbal light-based HMI has the potential to effectively alert passengers.

Our proposed HMI for passengers in automated buses aims to reduce anxiety by delivering information about the vehicle's behavior and status through ambient light, leveraging passengers' peripheral vision while they engage in personal tasks. This approach is anticipated to enhance the societal acceptance of autonomous buses, paving the way for their widespread utilization in our communities.

2 RELATED WORKS

Renate used a HUD to display various objects to dispel anxiety about self-driving vehicles and concluded that the display of a miniature world was most acceptable for the participants [1].

Harada concluded that the factors causing anxiety in experiments with personal mobility vehicles were approaching pedestrians and sudden changes in vehicle behavior and proposed the visual presentation of vehicle behavior information to participants [2].

Wickens proposed that peripheral vision can be used to receive information while human's central vision was occupied [3].

Gonçalves concluded in their experiments that an ambient-light-based HMI conveying takeover requests to the driver was as reliable and effective as auditory-based HMIs [4].

3 THEORY

In our study, we plan to conduct experiments in both real space and virtual reality space, recognizing that each environment has its distinct advantages and drawbacks (Table 1.).

Given these considerations, we aim to capitalize on the strengths of both environments by conducting experiments in each. Real experiments provide a platform for authentic sensory experiences, enabling precise evaluations. On the other hand, VR experiments offer the flexibility to explore potentially hazardous situations in a controlled setting. The findings gained from each experiment will be comprehensively evaluated based on the cumulative results and insights will be integrated.

* ogino.satoru@aist.go.jp

Table 1. The pros and cons of experiments in real and VR space

	Environment	
	Real space experiment	VR space experiment
Pros	Participants can engage with realistic sensations	- Participants can safely experiment with dangerous situations. - Experimental setup is relatively straightforward. - Less stringent constraints compared to real traffic laws. - Facilitates efficient testing of various scenarios.
Cons	- Experiments with potential danger are not feasible. - Difficulty in replicating the same situations for direct comparison. - Constraints imposed by rules and regulations.	- Challenges in conveying a truly realistic experience to participants. - Possibility of participants experiencing discomfort or motion sickness.

4 APPARATUS

As shown in Figure 1, a system to conduct real space experiments is constructed. The system aims to deliver appropriate information to passengers based on the driving information from an automated bus, GNSS, and acceleration sensor data.

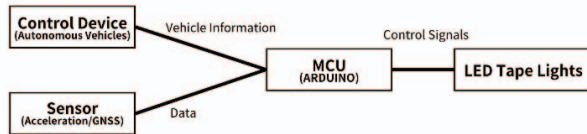


Figure 1: System Configuration



Figure 2: Ambient lighting in real space



Figure 3: Ambient lighting in VR

To evaluate the effectiveness of the ambient lighting, a bus route in Hitachi City, Ibaraki Prefecture, is reproduced in VR space. In both real space and VR space, the triggers to light different colors of bands are based on the bus's speed, acceleration, and positional information. The ambient lighting is developed in VR space as shown in Figure 2,3 and 4.



Figure 4: Rendering Images & VR Glasses

5 DISCUSSION

The current system design employs a three-color lighting scheme and blinking pattern to communicate information. Turquoise blue signifies normal operation, while orange indicates caution, and red signals danger. The blinking pattern includes both regular flashes and flowing flashes to convey caution and danger, with the primary goal of capturing passengers' attention. The effectiveness of such lighting as a means of communicating information in a driverless bus needs to be thoroughly examined. This examination is crucial to determine whether the system successfully conveys the vehicle's intentions to passengers. The results will contribute valuable insights to the enhancement of the HMI in driverless vehicles, providing better user experience to its passengers.

In the research demo, we will present our apparatus in both VR and real space. On the day of the conference, we will showcase how the system functions in each space, offering attendees a comprehensive understanding of our research. The establishment of such experiment environments allows us to conduct effective and efficient evaluations.

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